

ALIYA ALSAFA

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EDUCATION

Stanford University

M.S. Computer Science, Artificial Intelligence track (coterminal)

Palo Alto, CA

Expected Jun 2027

B.S. Computer Science, Systems track; Minor in Music, Piano Performance · GPA 3.70

Jun 2026

EXPERIENCE

Chainguard | Security Engineering Intern, Product Security · Remote

Summer 2026

- Designed a test corpus of pull requests seeded with known vulnerability classes, giving the benchmark a ground truth to measure against, and wrote the test suite and harness that run detectors over it.
- Researched and implemented 4 single-agent and multi-agent architectures against Claude, GPT, and Gemini models on Google Vertex AI; targeting 7–8 by the end of the internship.
- Exposed Semgrep and CodeQL to the agents as MCP tools while also running them as standalone baselines, separating what the models add from what static analysis already catches.
- Scored each architecture on vulnerabilities found, false positive rate, and cost per run, including whether its line-level comments were specific enough for a reviewer to act on.

Stanford University | Research Assistant, Biomechanical Modeling of Piano Performance

2025 – Present

- Training reinforcement learning policies in PyTorch to drive a physics-based model of piano-playing motion in NVIDIA Isaac Gym, over motion data retargeted from Blender, to study how simulated muscle activation informs piano performance technique.

Stanford Empirical Security Research Group | Undergraduate Researcher, advised by Zakir Durumeric

Apr 2025 – Jun 2026

- Built Rust modules in the Retina network analysis framework to offload flow-rule installation onto commodity NICs (DPDK, Mellanox).
- Contributed Retina's functional test harness (Python runner over pcap fixtures and expected output, plus a Rust test crate), merged upstream after code review.
- Designed experiments measuring NIC performance (rule latency, drop counters, RSS distribution) under high-throughput loads.
- Joined through CURIS; presented the poster “Reducing Host Load Via Commodity NIC Offloading” at the ESRG summer session and completed it as a CS 191W senior project, now a CoNEXT 2026 submission.

Redis | Security Engineering Intern · Remote

Summer 2024

- Reproduced HackerOne bug bounty findings with penetration testing techniques and wrote an internal offensive-testing guide for developers.
- Built a Python tool to find and report abandoned corporate DNS records, reducing cloud attack surface; containerized and deployed it to AWS ECS.
- Wrote custom Wiz frameworks to scan the cloud environment against CIS Ubuntu Benchmarks; integrated DoControl into corporate SaaS apps.

SKILLS

Languages: Python, Rust, Go, C/C++, Java, SQL, Terraform

AI & Agents: Claude, OpenAI, and Gemini APIs via Vertex AI; MCP; agent orchestration; PyTorch; Isaac Gym; pytest eval harnesses

Security & Cloud: Semgrep, CodeQL, Wiz, DoControl, BlinkOps, Wireshark, HackerOne, AWS, GCP, Docker, Linux/Unix

Tools: Git, DPDK, Jira, Sumo Logic

AWARDS & ACTIVITIES

TryHackMe Campus Ambassador, Stanford — organize cybersecurity events for students on campus.

Juilliard Pre-College, diploma in classical piano (2015–2022), performing internationally; Lang Lang International Music Foundation Young Scholar (2018–present).

Stanford Applied Cyber; Women in Computer Science; Executive Board, Stanford Piano Society and Stanford XTRM Dance.